# MTH2008 Real Analysis

## January 2023

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SOLUTIONS.

1 (a) (i) $S$ is not open since $0$ is not an interior point.

$S$ is not closed since $-1$ is a limit point that is not in the set.

(ii) $S^o = (-\infty, -1)$, $S^c = [-1, 0) \cup (0, \infty)$

(iii) Limit points : $(-\infty, -1]$

(iv) $\partial S = \{-1, 0\}$.

(Seen similar)

[8 Marks]

(b) Using the $\varepsilon$-$\delta$ definition of a limit prove that

$$\lim_{x \to 0} x \cos\left(\frac{1}{x^3}\right) = 0.$$

Solution. (Seen similar)

$f(x) = x\cos\left(\frac{1}{x^3}\right)$ is defined in a deleted neighborhood of 0 and for all $x \neq 0$,

$$\left| x\cos\left(\frac{1}{x^3}\right) \right| \leq |x|.$$

Thus, for all $\varepsilon > 0$ there exists $\delta = \varepsilon$ such that if $0 < |x| < \delta$, then

$$\left| x\cos\left(\frac{1}{x^3}\right) \right| \leq |x| < \delta = \varepsilon.$$

1 (c) Prove, using $\varepsilon$ and $\delta$, that the function

$f: \mathbb{R}_{\geq 0} \to \mathbb{R}$ defined by

$$f(x) = \sqrt{8x} - 4$$

is continuous from the right at 2.

Solution. (Seen similar)

Since $f(2) = 0$, we have to prove that

$$\lim_{x \to 2^+} f(x) = 0.$$

Fix $\varepsilon > 0$; then the number

$\delta = \frac{(4+\varepsilon)^2}{8} - 2$ satisfies $\delta = \frac{\varepsilon^2}{8} + \varepsilon > 0$.

Thus, if $0 < x - 2 < \delta$, then

$$x < \delta + 2 = \frac{(4+\varepsilon)^2}{8} \Rightarrow 8x < (4+\varepsilon)^2$$

$$\Rightarrow \sqrt{8x} < 4 + \varepsilon,$$

and so $|\sqrt{8x} - 4| = \sqrt{8x} - 4 < \varepsilon$ ∎

(Seen similar but more difficult) (10 Marks)

1 (d) (Unseen) [4]

Find $\lim_{n\to\infty} s_n$ for the following sequence, justifying your answer using the $\varepsilon$-$N$ definition of a limit:

$$s_n = \frac{n}{n+\sqrt{n}}.$$

Solution.

By inspection the limit is 1:

For large $n$, $\sqrt{n} \ll n$, hence

$$\frac{n}{n+\sqrt{n}} \approx \frac{n}{n} = 1.$$

To show this rigorously, we have

$$|s_n - 1| = \left| \frac{n}{n+\sqrt{n}} - 1 \right| = \left| \frac{-\sqrt{n}}{n+\sqrt{n}} \right| =$$

$$= \left| \frac{-1}{\sqrt{n}+1} \right| = \frac{1}{\sqrt{n}+1} \leq$$

$$\leq \frac{1}{\sqrt{N}+1} < \frac{1}{\frac{1}{\varepsilon}+1} = \frac{\varepsilon}{\varepsilon+1} < \varepsilon,$$

for all $n \geq N$ with $N > \frac{1}{\varepsilon^2}$.

(This is not the only possibility for $N$: any suitable $N$ is correct).

(e) Use Riemann sums to show that the function

$$f(x) = x^2 + 4x$$

is integrable on $[0,1]$ and hence determine the value of its definite integral.

Solution. (Seen similar)

Take a uniform partition $P_n$ of $[0,1]$ made up of the points $x_i = \frac{i}{n}$, $i = 0, \dots, n$.

Since $f$ is increasing on the interval, we have the lower sum

$$L(f, P_n) = \sum_{i=1}^{n} \left( \left( \frac{i-1}{n} \right)^2 + 4 \left( \frac{i-1}{n} \right) \right) \cdot \frac{1}{n} =$$

$$= \frac{1}{n} \sum_{i=1}^{n} \left( \frac{i^2 - 2i + 1}{n^2} + \frac{4i - 4}{n} \right) =$$

$$= \frac{1}{n^3} \sum_{i=1}^{n} i^2 - \frac{2}{n^3} \sum_{i=1}^{n} i + \frac{1}{n^3} \cdot n +$$

$$+ \frac{4}{n^2} \sum_{i=1}^{n} i - \frac{4}{n^2} \cdot n =$$

$$= \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} - \frac{2}{n^3} \cdot \frac{n(n+1)}{2} + \frac{1}{n^2} +$$

$$+ \frac{4}{n^2} \cdot \frac{n(n+1)}{2} - \frac{4}{n} =$$

$$= \frac{1}{n^3} \cdot \frac{n(2n^2+n+2n+1)}{6} - \frac{2}{n^3} \cdot \frac{n^2+n}{2} + \frac{1}{n^2} +$$

$$+ \frac{4}{n^2} \cdot \frac{n^2+n}{2} - \frac{4}{n}$$

$$= \frac{2n^3+3n^2+n}{6n^3} - \frac{2}{n^3} \cdot \frac{n^2+n}{2} + \frac{1}{n^2} +$$

$$+ \frac{4}{n^2} \cdot \frac{n^2+n}{2} - \frac{4}{n} =$$

$$= \frac{2n^2+3n+1}{6n^2} - \frac{2n+2}{2n^2_{(3)}} + \frac{1}{n^2_{(6)}} +$$

$$+ \frac{2n+2}{n_{(6n)}} - \frac{4}{n_{(6n)}} =$$

$$= \frac{2n^2+3n+1}{6n^2} - \frac{6n+6}{6n^2} + \frac{6}{6n^2} +$$

$$+ \frac{12n^2+12n}{6n^2} - \frac{24n}{6n^2} =$$

$$= \frac{2n^2+3n+1-6n-6+6+12n^2+12n-24n}{6n^2}$$

$$= \frac{14n^2}{6n^2} - \frac{15n}{6n^2} + \frac{1}{6n^2} =$$

$$= \frac{7}{3} - \frac{5}{2n} + \frac{1}{6n^2}$$

and the upper sum

$$U(f, P_n) = \sum_{i=1}^{n} \left( \left( \frac{i}{n} \right)^2 + 4\left( \frac{i}{n} \right) \right) \cdot \frac{1}{n}$$

$$= \frac{1}{n} \sum_{i=1}^{n} \left( \frac{i^2}{n^2} + \frac{4i}{n} \right) =$$

$$= \frac{1}{n^3} \sum_{i=1}^{n} i^2 + \frac{4}{n^2} \sum_{i=1}^{n} i =$$

$$= \frac{1}{n^3} \cdot \frac{2n^3 + 3n^2 + n}{6} + \frac{4}{n^2} \cdot \frac{n^2 + n}{2}$$

$$= \frac{2n^2 + 3n + 1 + 12n^2 + 12n}{6n^2}$$

$$= \frac{14n^2 + 15n + 1}{6n^2}$$

$$= \frac{7}{3} + \frac{5}{2n} + \frac{1}{6n^2}$$

We have $\sup_n L(f, P_n) = \frac{7}{3}$ and $\inf_n U(f, P_n) = \frac{7}{3}$.

Hence, since $\sup_n L(f, P_n) \leq \sup_P L(f, P)$ and $\inf_n U(f, P_n) \geq \inf_P U(f, P)$, we must have

$$\int_0^1 (x^2 + 4x)\, dx = \frac{7}{3}.$$

[2] (a)

Let $A$ be a nonempty subset of $\mathbb{R}$ bounded above. Set

$$B = \{-a \mid a \in A\}.$$

Show that $B$ is bounded below and that

$$\inf B = -\sup A. \qquad \text{(10 Marks)}$$

Solution. (Unseen)

Since $A$ is bounded above, there exists $m \in \mathbb{R}$ such that

$$\forall a \in A,\ a \leq m.$$

Hence $\forall a \in A$, we have $-m \leq -a$, which implies that for all $b \in B$ we have $-m \leq b$.

Thus, $B$ is bound below and therefore, $\inf B$ exists.

Let us now show that $\inf B = -\sup A$.

By the definition of $\sup A$,

$\forall a \in A,\ a \leq \sup A$, so

$\forall a \in A,\ -\sup A \leq -a$, or

$\forall b \in B,\ -\sup A \leq b$.

The definition of $\inf B$ implies

$-\sup A \leq \inf B$.

Next, we have

$\forall b \in B,\ \inf B \leq b$, which implies

$\forall b \in B,\ -b \leq -\inf B$, or

$\forall a \in A,\ a \leq -\inf B$

since $A = \{-b : b \in B\}$.

By the definition of $\sup A$ we get $\sup A \leq -\inf B$.

Combining this with $-\sup A \leq \inf B$, we deduce that

$\inf B = -\sup A$. ∎

[2] (b) (Unseen)

Let $\lim_{x \to x_0} f(x) = l$. Let us prove that $\lim_{x \to x_0} |f(x)| = |l|$. Let $\varepsilon > 0$. Then there exists $\delta > 0$ such that for any $x \in D$, $|f(x) - l| < \varepsilon$ provided $|x - x_0| < \delta$.

Using the inequality

$$||a| - |b|| \leq |a - b|$$

for any real numbers $a$ and $b$ we have that for any $x \in D$ such that $|x - x_0| < \delta$, we have

$$||f(x)| - |l|| \leq |f(x) - l| < \varepsilon.$$

This completes our proof.

2 (c) (Seen similar)

Fix $\varepsilon > 0$. Then for $\delta = \sqrt[3]{\frac{1}{\varepsilon}}$, we get that if $x > \delta$, then

$$\left| \frac{1}{x^3 + 2x + 1} \right| \leq \left| \frac{1}{x^3} \right| < \frac{1}{\delta^3} = \varepsilon$$

as we wanted to show.

3 (a) State the Mean Value Theorem.

Solution. (Bookwork)

If $f$ is continuous on the closed interval $[a, b]$ and differentiable on the open interval $(a, b)$, then

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

for some $c \in (a, b)$.

(ii) If $f$ is constant on $(a,b)$, then there exists $\alpha \in \mathbb{R}$ such that $f(x) = \alpha$ for all $x \in (a,b)$.

Then $f'(x) = 0$ for all $x \in (a,b)$.

Vice-versa, if $f'(x) = 0$ for all $x \in (a,b)$, we apply the Mean Value Theorem to the closed interval $[x_1, x_2]$ where $x_1, x_2$ are two elements of $(a,b)$ and we get that $f(x_1) = f(x_2)$.

This argument holds for all $x_1, x_2 \in (a,b)$, hence $f$ is constant in $(a,b)$.

(Unseen)

3 (Unseen)

(b) let

$$\lim_{n\to\infty} x_{2n} = \lim_{n\to\infty} x_{2n+1} = l$$

and let us prove that

$$\lim_{n\to\infty} x_n = l.$$

Let $\varepsilon > 0$. There exist $N_0 \geq 1$ and $N_1 \geq 1$ such that

$|x_{2n} - l| < \varepsilon$, for all $n \geq N_0$,

$|x_{2n+1} - l| < \varepsilon$, for all $n \geq N_1$.

Set $N = \max\{2N_0, 2N_1 + 1\}$. Let $n \geq N$.

If $n = 2k$, then we have $k \geq N_0$ since $n \geq N \geq 2N_0$.

Using the above inequalities we get $|x_{2k} - l| < \varepsilon$, or $|x_n - l| < \varepsilon$. A similar argument when $n$ is odd will yield the same inequality.

Therefore, $|x_n - l| < \varepsilon$, for any $n \geq N$.

This proves that $\{x_n\}$ is convergent.

[4] (a) (Bookwork)

Let $f : [a,b] \to \mathbb{R}$ be a bounded function on the interval $[a,b]$, and $P = (x_0, x_1, \ldots, x_n)$ be a partition of $[a,b]$. ~~Define~~ Give the definition of the lower and upper Riemann sum of $f$ with respect to the partition $P$.

Solution. (Bookwork)

The lower Riemann sum of $f$ with respect to the partition $P$ is

$$L(f;P) = \sum_{i=1}^{n} m_i \Delta x_i ,$$

the upper Riemann sum of $f$ with respect to the partition $P$ is

$$U(f;P) = \sum_{i=1}^{n} M_i \Delta x_i$$

where $\Delta x_i = x_i - x_{i-1}$, $M_i = \sup_{x \in [x_{i-1}, x_i]} f(x)$,

$m_i = \inf_{x \in [x_{i-1}, x_i]} f(x)$.

4 (b)

Prove, using an $\varepsilon$-$\delta$ argument, that the function $f: \mathbb{R}^2 \to \mathbb{R}$, given by

$$f(x) = \begin{cases} \dfrac{x_1^2 x_2^2}{x_1^2 + x_2^2}, & \text{if } x \neq 0, \\ 0, & \text{if } x = 0, \end{cases}$$

is continuous at $x = 0$.

Solution.

We have

$$\|f(x) - f(0)\| = \frac{x_1^2 x_2^2}{x_1^2 + x_2^2}$$

$$\leq \frac{(x_1^2 + x_2^2)^2}{x_1^2 + x_2^2}$$

$$= x_1^2 + x_2^2$$

$$< \delta^2 < \varepsilon$$

holds for all $0 < \|x - 0\| = \sqrt{x_1^2 + x_2^2} < \delta$ with $\delta = \sqrt{\varepsilon}$.

Hence, $\lim_{x \to 0} f(x) = f(0)$ and $f$ is continuous at 0.

(Seen similar)

4 (a) (Seen similar)

(i) $\dfrac{\partial f_1}{\partial x_1} = -x_2 \sin(x_1 x_2)$

$\dfrac{\partial f_2}{\partial x_1} = x_2 \cos(x_1 x_2)$

$\dfrac{\partial f_1}{\partial x_2} = -x_1 \sin(x_1 x_2)$

$\dfrac{\partial f_2}{\partial x_2} = x_1 \cos(x_1 x_2)$

where $f_1(x_1, x_2) = \cos(x_1 x_2)$,

$f_2(x_1, x_2) = \sin(x_1 x_2)$.

The derivative is

$$\begin{pmatrix} -x_2 \sin(x_1 x_2) & -x_1 \sin(x_1 x_2) \\ x_2 \cos(x_1 x_2) & x_1 \cos(x_1 x_2) \end{pmatrix}.$$

(ii) The function $f$ is differentiable at $(x_1, x_2)^T = (\pi, 1)^T$ because its partial derivatives are continuous (at this point).

[10 Marks].

3 (c) (Bookwork)

$f: \mathbb{R}^n \to \mathbb{R}$, $f(x) = \|x\|^2$, $p \in \mathbb{R}^n$

Let us look at the limit

$$\lim_{h \to 0} \frac{\|f(p+h) - f(p) - Df(p)h\|}{\|h\|}$$

$$f(x) = \|x\|^2 = x_1^2 + x_2^2 + \cdots + x_n^2$$

$$f(p, h) = \|p + h\|^2 =$$

$$= (p_1 + h)^2 + (p_2 + h)^2 + \cdots + (p_n + h)^2$$

$$= \|p\|^2 + 2\langle p, h \rangle + \|h\|^2$$

We have the partial derivatives

$$Df(x) = (2x_1, 2x_2, \ldots, 2x_n)$$

and at $p$ we have

$$Df(p) = 2(p_1, p_2, \ldots, p_n)$$

and so

$$Df(p)\cdot h = 2(p_1, p_2, \ldots, p_n)\cdot h =$$

$$= 2\langle p, h\rangle$$

So, the limit is

$$\lim_{h\to 0} \frac{\|\|p\|^2 + 2\langle p, h\rangle + \|h\|^2 - \|p\|^2 - 2\langle p, h\rangle\|}{\|h\|}$$

$$= \lim_{h\to 0} \|h\| = 0.$$